PROJECT BLUEPRINT & SOLUTION ARCHITECTURE

RO Water Service Provider Website + AI-Powered CRM

Unified cold-calling & online-lead management with an autonomous AI telecalling engine, smart lead filtering, and seamless human hand-off.

Next.js	**PostgreSQL**	**Twilio + GPT-4o**	**50+ agents**
Web + App	+ Prisma ORM	Realtime Voice AI	5,000+ calls / day

DOCUMENT	PREPARED FOR	DATE	STATUS
Solution Blueprint v1.0	RO Water Service Provider	October 2026	Awaiting Client Approval

Contents

1. Executive Summary
2. Business Objectives & Success Metrics
3. High-Level Solution Architecture
4. Technology Stack
5. Module 1 — Public Website
6. Module 2 — Customer Portal
7. Module 3 — CRM & Unified Lead Inbox
8. Module 4 — AI Telecalling Engine
9. AI Voice Agent — Call Flow & Scripts
10. Module 5 — AI Automation Workflows
11. Human Hand-off Mechanism
12. Lead Sources & Data Ingestion
13. Database Schema (Prisma)
14. Analytics, Dashboards & Reporting
15. Security, Compliance & DND (TRAI)
16. Scalability & Infrastructure
17. Implementation Roadmap
18. Team, Timeline & Budget
19. Deliverables & Next Steps
20. Risks & Assumptions

1. Executive Summary

This blueprint defines a single, unified platform that combines a customer-facing **RO water service website** with an internal **AI-powered CRM**. It replaces fragmented cold-calling spreadsheets and scattered online leads with one automated pipeline: every lead — whether uploaded by the team or captured from digital marketing — is enriched, called by an AI voice agent, qualified, filtered into the correct segment, and routed to a human only when the customer asks for one or as needed.

1	24×7	100%
Unified Lead Panel	AI Telecalling	Automatic Follow-ups

The Problem Today

- Cold-calling data lives in spreadsheets; online leads live in ad dashboards — no single source of truth.
- Human telecallers are expensive, inconsistent, and cannot scale past a few hundred calls a day.
- Leads go cold because follow-ups are manual and forgettable.
- No reliable qualification/filtering; sales agents waste time on unqualified contacts.
- No visibility into which channel produces real, paying customers.

The Solution

A **Next.js + PostgreSQL + Prisma** platform with a **Twilio + OpenAI GPT-4o Realtime** voice engine that autonomously dials leads, holds natural conversations, qualifies intent (new install, AMC, repair, filter/part replacement), answers FAQs, detects a request for a human, and writes every outcome back into a single CRM database.

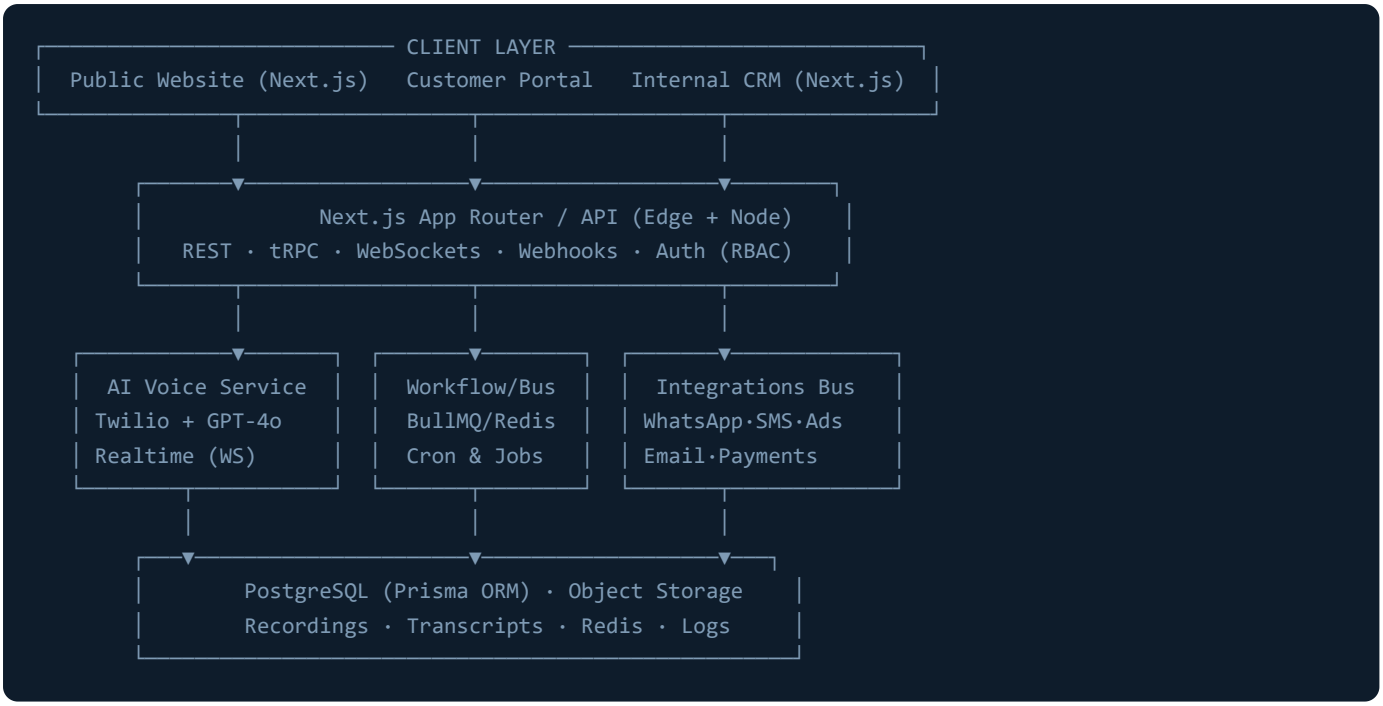
Core promise: The owner sees **every cold-call and online lead in one panel**, fully automated by AI, with clean segmentation and an instant human escalation path — no lead is ever lost.

2. Business Objectives & Success Metrics

Objective	How We Deliver It	KPI / Target
One panel for all leads	Unified CRM ingest from uploads, website, ads, WhatsApp, missed calls	100% of leads in one database
AI does all telecalling	Autonomous outbound voice agent with retry logic	≥ 90% of dials handled end-to-end by AI
Filter/qualify new clients	AI scoring + segmentation into warm/cold/not-interested	≥ 35% qualified-lead rate
Human hand-off on request	Live warm transfer + callback ticket	< 15s hand-off latency
Automate the full process	Workflow engine: enrich → dial → follow-up → assign	0 manual follow-ups required
Scale	Concurrent AI calls + 50+ human seats	5,000+ calls/day
Revenue growth	Channel attribution + conversion analytics	+30% qualified pipeline in 90 days

3. High-Level Solution Architecture

1. **Lead Sources** — Team CSV upload, Website forms & chatbot, AI-generated, Google/Facebook Ads, WhatsApp, missed-call, walk-in.
2. **Data Ingestion & Enrichment** — Dedup, validation, DND scrub, geo/language tagging, intent tagging.
3. **AI Telecalling Engine** — Twilio Programmable Voice + OpenAI GPT-4o Realtime → natural conversation, qualification, objection handling.
4. **CRM Processing** — Auto scoring, segmentation, tags, call recording, transcript, sentiment.
5. **Routing** — Qualified → booking/AMC; "talk to human" → live warm transfer or callback ticket; not interested → nurture.
6. **Automation** — Follow-up cadences (SMS/WhatsApp/call), reminders, review requests, renewal notices.
7. **Analytics** — Funnel, agent leaderboard, channel ROI, AI vs human performance.


CLIENT LAYER
Public Website (Next.js)
Customer Portal
Internal CRM (Next.js)
Next.js App Router / API (Edge + Node)
REST · tRPC · WebSockets · Webhooks · Auth (RBAC)
AI Voice Service
Twilio + GPT-4o
Realtime (WS)
Workflow/Bus
BullMQ/Redis
Cron & Jobs
Integrations Bus
WhatsApp·SMS·Ads
Email·Payments
PostgreSQL (Prisma ORM) · Object Storage
Recordings · Transcripts · Redis · Logs

4. Technology Stack

Layer	Technology	Why
Frontend (Website + CRM + Portal)	Next.js (App Router), React, TypeScript, Tailwind CSS, shadcn/ui	SSR/SEO for website, fast SPA-like CRM, one codebase
Backend / API	Next.js Route Handlers + tRPC, Node.js	Type-safe end-to-end, minimal overhead
Database	PostgreSQL 16 + Prisma ORM	Relational integrity for leads/calls/services
Cache / Queues	Redis + BullMQ	Dialer queues, retries, rate limits, real-time
Telephony	Twilio Programmable Voice + Media Streams	Scalable outbound/inbound + SIP trunk
AI Conversation	OpenAI GPT-4o Realtime API	Low-latency natural voice, barge-in, multilingual
Speech	OpenAI TTS/Whisper or Deepgram + ElevenLabs (Hindi/English)	Accurate STT/TTS, Indian-language support
Messaging	Twilio WhatsApp Business API, Twilio SMS, SendGrid	Automated follow-ups & confirmations
Payments	Razorpay / Stripe	AMC renewals, service payments
Maps/Routing	Google Maps Platform	Technician scheduling & service-area logic
Storage	S3-compatible object storage	Recordings, transcripts, documents
Auth	NextAuth / Auth.js (JWT + RBAC)	Roles: owner, manager, agent, technician
Monitoring	Sentry, Grafana + Prometheus, structured logs	Reliability & call-quality tracking
Deployment	Docker + Vercel / AWS (ECS) or Hostinger VPS + Nginx	Flexible, cost-effective scale

5. Module 1 — Public Website (Lead Generation)

Marketing website optimized for SEO and conversion for an RO water provider.

Pages

- Home (USPs, trust, CTA)
- Services: New RO Installation, AMC, Repair, Filter/Parts, Water Testing
- Products / RO Models
- Pricing & AMC plans
- Service areas / locations
- Blog & water-health guides (SEO)
- About, Contact, FAQs

Conversion Features

- Multi-step "Book a Service" form
- Instant quote calculator (RO type, area)
- Click-to-call & WhatsApp button
- AI website chatbot → creates CRM lead
- Missed-call auto lead capture
- Landing pages per ad campaign (UTM)
- Reviews & testimonials, trust badges

SEO & Speed: Next.js SSR/ISR, structured data (LocalBusiness, Service), sitemap, Core Web Vitals ≥ 90, PWA-ready.

6. Module 2 — Customer Portal

Self-service area for existing customers (login via mobile + OTP).

Feature	Description
My Devices / AMC	Registered RO units, install date, AMC status & expiry
Book / Reschedule Service	Choose date-slot, technician auto-assign by area
Service History	Past visits, parts replaced, invoices
Live Tracking	Technician ETA & status
Payments & Invoices	Online payment, download GST invoice
Renewals & Reminders	AMC/filter renewal auto-notices
Support / Chat	Raise ticket, AI chatbot, callback request
Referral	Refer & earn credits

7. Module 3 — CRM & Unified Lead Inbox

The single control panel where **cold-calling and online leads** converge.

Lead Inbox (Kanban + Table)

- Pipeline stages: New | AI Calling | Qualified | Interested | Human Follow-up | Converted | Nurture | Lost/DND

- Source badges: Cold DB Upload · Website · Google Ads · Meta Ads · WhatsApp · Missed Call · Referral
- Filters: source, city, language, RO type, intent, AI score, last-call outcome, assignee
- Timeline per lead: calls, transcripts, notes, messages, bookings, payments

Lead 360° Profile

- Contact details & preferred language
- Multiple RO units / address
- AI qualification score (0–100)
- Intent: New · AMC · Repair · Parts
- Best time to call, consent/DND status
- Full conversation history & recordings

Agent Workspace

- Assigned leads queue & daily targets
- Click-to-call (browser softphone)
- AI-generated call summary & next action
- Quick SMS/WhatsApp templates
- Disposition & follow-up scheduling
- Booking creation inline

Roles & Permissions (RBAC): Owner (full), Manager (all leads + reports), Team Lead (team leads + assign), Agent (own leads), Technician (job sheet only), Accountant (billing).

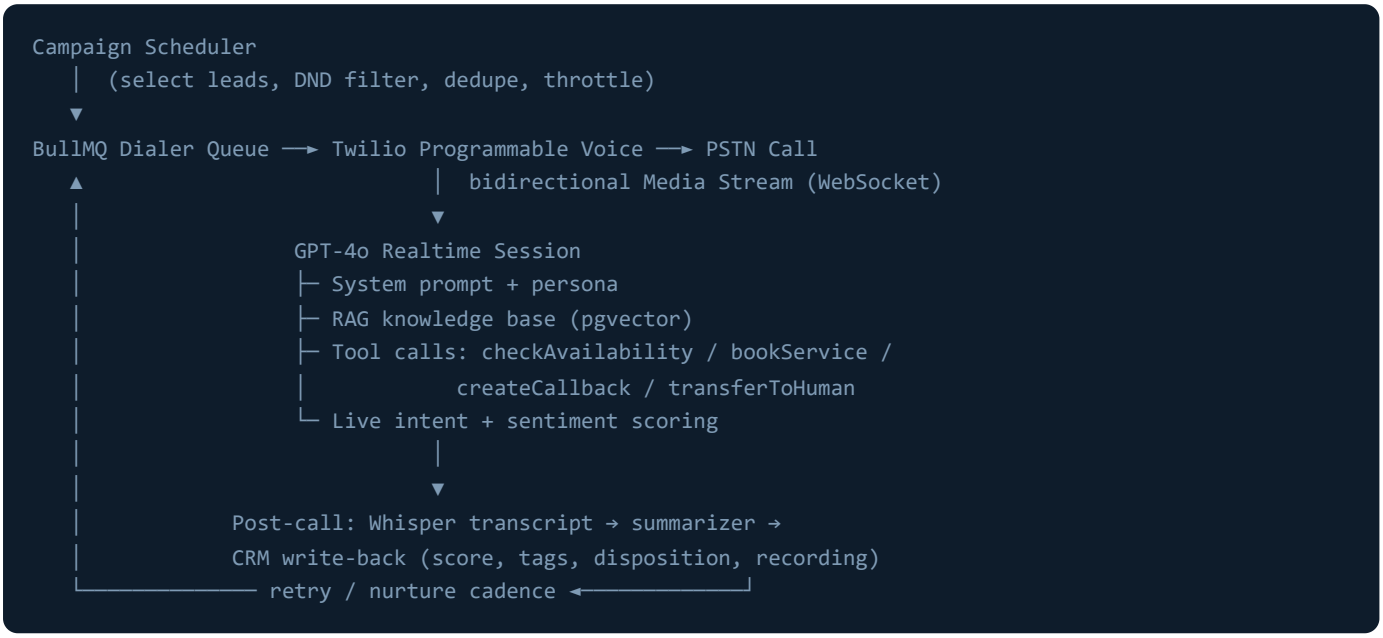
8. Module 4 — AI Telecalling Engine

Autonomous outbound + inbound voice automation built on **Twilio Programmable Voice** and the **OpenAI GPT-4o Realtime API**.

Core Capabilities

- **Auto-dialing:** Preview, Progressive & Predictive modes with configurable concurrency per campaign.
- **Natural conversation:** Realtime speech-to-speech, barge-in, natural pauses, dialect-aware.
- **Multilingual:** Hindi, English, Hinglish (extensible to regional languages).
- **Qualification engine:** Asks scripted + dynamic questions, extracts intent, budget, location, RO type, urgency.
- **Knowledge base (RAG):** Answers pricing, AMC terms, service areas from an updatable knowledge base.
- **Objection handling:** Trained responses to "not interested", "too costly", "already have RO".
- **Human detection:** Recognizes "talk to a person / human / agent" keywords & sentiment → triggers hand-off.
- **Compliance guardrails:** Honest AI disclosure, DND/TRAI checks, call-recording consent.
- **Auto follow-up:** No-answer/busy → scheduled retries; positive → confirmation SMS/WhatsApp.
- **Post-call AI:** Transcription → summary → sentiment → score → next best action.

Engine Architecture



9. AI Voice Agent — Call Flow & Sample Scripts

1. **Greeting & disclosure:** "Namaste, main [Company] se ek AI assistant hoon..."
2. **Purpose:** Confirm interest & qualify need (new RO, service, AMC, repair).
3. **Qualify:** RO present? Brand age? Problem? City/area? Best time?
4. **Handle objections** & present relevant offer / plan.
5. **Detect human request** → `transferToHuman` or callback ticket.
6. **Close/next step:** Book service, send quote on WhatsApp, or schedule follow-up.
7. **Wrap-up:** Confirm details, thank, log outcome & score in CRM.

Sample Script — Cold Call (Hinglish)

```
AI: Namaste! Main [Company] se ek AI assistant hoon. Aapke area mein
    RO water purifier service ke liye call kar raha hoon. 2 minute hai?
Customer: Haan boliye.
AI: Bahut badhiya. Kya ghar ya office mein RO purifier lagana chahte hain,
    ya existing purifier ki servicing karani hai?
Customer: Servicing karani hai.
AI: Theek hai. Aapka purifier kitne saal purana hai aur last service
    kab hui thi?
Customer: 2 saal, 6 mahine pehle.
AI: Samajh gaya. Filters change + general service recommend karta hoon.
    Aap kaunse area se hain, taaki main nearest technician confirm kar saku?
Customer: [Area]
AI: Perfect. Main aapke liye ek service slot book kar sakta hoon.
    Kal shaam 5 baje ka slot available hai — chalega?
Customer: Mujhe detail chahiye, kisi insaan se baat karni hai.
AI: Bilkul! Main abhi aapko hamare service expert se connect kar raha hoon.
    Ek moment please… [→ live transfer]
```

Qualification & Scoring Model

Signal	Weight	Source
Explicit intent (new/AMC/repair)	30	Conversation
Service area match	20	Geo / address
Positive sentiment / engagement	20	Realtime sentiment
Budget / offer acceptance	15	Conversation
Urgency / timeline	15	Conversation

Score ≥ 70 → Qualified (auto-book/assign). 40–69 → Warm nurture. < 40 → Nurture/Lost. Any "human" request → immediate hand-off regardless of score.

10. Module 5 — AI Automation Workflows

A visual workflow engine (trigger → condition → action) automates every step without human effort.

Workflow	Trigger	Automated Actions
New Lead Nurture	Lead created (upload/web/ads)	Validate → DND scrub → dedupe → enrich → enqueue AI call
AI Cold Call Cadence	Enqueued lead	Call → if no answer retry 3× (spaced) → SMS/WhatsApp if still no
Auto-Qualification	Call ends	Transcribe → score → tag → set stage → notify assignee
Human Hand-off	"talk to human" / high-value	Live warm transfer; if unavailable → callback ticket + notify
Booking Confirmation	Service booked	WhatsApp + SMS confirmation → assign technician → calendar event
Post-Service Follow-up	Job completed	Feedback request → review link → upsell AMC
AMC / Filter Renewal	X days before expiry	Reminder call → WhatsApp → payment link → renewal
Re-engagement	Lost/nurture after N days	New AI call with fresh offer
Review & Referral	Payment received / 5-star	Send review + referral link

AI everywhere: Lead scoring, call summaries, reply suggestions, spam detection, transcript search, and predictive best-time-to-call are all AI-driven. The owner only supervises.

11. Human Hand-off Mechanism

When a customer asks for a human (or an edge case is detected), the system escalates instantly.

Path A — Live Warm Transfer

- AI detects trigger phrase / sentiment
- Looks up available agent (skill + language + area)
- Uses Twilio <Dial> to bridge the call
- Passes a spoken/written context summary to agent
- Target: < 15s, call not dropped

Path B — Callback Ticket

- No agent available / outside hours
- AI collects preferred time & reason
- Creates high-priority ticket, assigns to agent
- Sends confirmation SMS/WhatsApp to customer
- Agent calls back within SLA

Trigger Phrases (configurable)

```
"talk to human", "agent se baat", "real person", "manager", "insaan", "customer care", "kisi aadmi se", sentiment = frustrated/angry, or lead score ≥ 90.
```

12. Lead Sources & Data Ingestion

Source	Method	Automation
Team CSV/Excel upload	Upload wizard + column mapper	Parse → dedupe → DND scrub → enqueue
Website forms / chatbot	Next.js API → CRM	Instant lead + AI call trigger
Google / Meta Ads	Webhooks + UTM capture	Tag channel → attribute ROI
WhatsApp enquiries	Twilio WhatsApp API	Auto-reply → create lead → AI call
Missed calls	Twilio missed-call webhook	Auto SMS + callback lead
Justdial / IndiaMART / 99acres	Lead APIs/email parser	Normalize → CRM
Walk-in / referral	Manual quick-add	Same pipeline

Dedup logic: match on normalized phone; merge duplicates; keep source trail. **Validation:** E.164 phone format, valid Indian mobile, TLD/email checks.

13. Database Schema (Prisma — Core Models)

```
model User        { id String @id @default(cuid())  name String
                    email String @unique  role Role  phone String?
                    leads Lead[]  calls Call[]  createdAt DateTime @default(now()) }

model Lead        { id String @id @default(cuid())
                    name String  phone String @unique  email String?
                    city String?  area String?  language String? @default("hi")
                    source Source  status LeadStatus @default(NEW)
                    intent Intent?  score Int?  dnd Boolean @default(false)
                    consent Boolean @default(false)
                    assignedToId String?  assignedTo User? @relation(fields:[assignedToId],references:[id])
                    ownerId String?  activities Activity[]  calls Call[]
                    createdAt DateTime @default(now())  updatedAt DateTime @updatedAt }

model Call        { id String @id @default(cuid())  leadId String
                    lead Lead @relation(fields:[leadId],references:[id])
                    direction Direction  status CallStatus
                    duration Int?  recordingUrl String?  transcript String?
                    summary String?  sentiment String?  disposition String?
                    aiScore Int?  handledByAi Boolean @default(true)
                    agentId String?  agent User? @relation(fields:[agentId],references:[id])
                    startedAt DateTime @default(now())  endedAt DateTime? }

model Activity    { id String @id @default(cuid())  leadId String
                    lead Lead @relation(fields:[leadId],references:[id])
                    type String  note String?  meta Json?  createdAt DateTime @default(now()) }

model ServiceJob  { id String @id @default(cuid())  leadId String?  customerId String?
                    type ServiceType  status JobStatus  scheduledAt DateTime?
                    technicianId String?  address String?  amount Decimal?
                    invoiceUrl String?  createdAt DateTime @default(now()) }

model ROKnowledge { id String @id @default(cuid())  title String
                    content String  embedding Unsupported("vector")? }

enum Role        { OWNER MANAGER TEAM_LEAD AGENT TECHNICIAN ACCOUNTANT }
enum LeadStatus  { NEW AI_CALLING QUALIFIED INTERESTED HUMAN_FOLLOWUP CONVERTED NURTURE LOST DND }
enum Source      { COLD_UPLOAD WEBSITE GOOGLE_ADS META_ADS WHATSAPP MISSED_CALL REFERRAL WALKIN PARTNER }
enum Intent      { NEW_INSTALL AMC REPAIR FILTER_PARTS WATER_TEST OTHER }
enum CallStatus  { QUEUED RINGING IN_PROGRESS COMPLETED NO_ANSWER BUSY FAILED }
enum Direction   { OUTBOUND INBOUND }
```

Supporting models: `Campaign`, `FollowUp`, `Message`, `Payment`, `Ticket`, `AuditLog`, `WebhookEvent`. pgvector used for the AI knowledge base (RAG).

14. Analytics, Dashboards & Reporting

Owner Dashboard

- Leads today: cold vs online
- Calls made / connected / qualified
- Conversion funnel & revenue
- AI vs human performance
- Channel ROI (Ads → sales)

Manager / Agent View

- Agent leaderboard & activity
- Call recordings + transcripts
- Follow-up & SLA compliance
- Service & AMC renewal tracking
- Exportable reports (CSV/PDF)

Funnel: Lead → Contacted → Qualified → Booked → Paid

AI metrics: Connect rate, avg talk time, sentiment, hand-off rate

Attribution: cost per qualified lead by source

15. Security, Compliance & DND (TRAI)

- **TRAI/DND compliance:** scrub numbers against DND registry; honor consent flags; calling-hour restrictions.
- **AI disclosure:** agent transparently identifies as AI (configurable by region).
- **Call recording consent:** announcement + configurable opt-in.
- **Data protection:** DPDP Act (India) alignment — encryption at rest/in transit, data retention policy, PII masking.
- **Access control:** RBAC, 2FA for admins, audit logs of every lead/call action.
- **Secrets & keys:** Twilio/OpenAI keys in a secrets manager; webhook signature verification.
- **Backups:** automated daily DB backups + point-in-time recovery.

Note: Outbound calling regulations vary by region. Final calling strategy (consent, timing, disclosure) will be confirmed with legal/compliance before go-live.

16. Scalability & Infrastructure (5,000+ calls/day)

Concern	Approach
Concurrent AI calls	Horizontal workers; Twilio concurrency tiers; configurable dial rate
Queue management	Redis + BullMQ with priorities, retry/backoff, dead-letter queue
Database load	Connection pooling (PgBouncer), read replicas, partitioning of call/activity tables
Latency	GPT-4o Realtime over WebSocket; edge routing; regional Twilio endpoints
50+ human seats	Softphone via Twilio Client; presence & skill-based routing
Observability	Call-quality dashboards, error tracking, alerting

17. Implementation Roadmap

Phase	Duration	Scope
Phase 0 — Discovery	1 week	Finalize flows, scripts, compliance, KPIs, integrations
Phase 1 — Foundation	2 weeks	Next.js + PostgreSQL + Prisma setup, auth, RBAC, base CRM schema
Phase 2 — CRM Core	3 weeks	Lead inbox, CSV upload, dedupe/DND, pipeline, timeline, agent workspace
Phase 3 — AI Voice Engine	4 weeks	Twilio + GPT-4o Realtime, dialer, qualification, RAG, post-call AI
Phase 4 — Human Hand-off & Automation	3 weeks	Live transfer, callback tickets, workflow engine, follow-up cadences
Phase 5 — Website & Portal	3 weeks	Public site, SEO, forms, chatbot, customer portal, payments
Phase 6 — Analytics & Integrations	2 weeks	Dashboards, ad/WhatsApp integrations, reporting
Phase 7 — QA, Load & Go-Live	2 weeks	Testing, 5k-call load test, training, deployment, hypercare

Total: ~20 weeks (≈5 months) for full scope. A lean MVP (CRM + AI calling + hand-off) can go live in ~8–10 weeks.

18. Team, Timeline & Budget

Recommended Team

- 1 × Solution Architect / Tech Lead
- 2 × Full-stack Engineers (Next.js + Prisma)
- 1 × AI/Voice Engineer (Twilio + GPT-4o Realtime)
- 1 × Frontend/UI Engineer

- 1 × QA Engineer
- 1 × DevOps (part-time)
- 1 × BA/PM (part-time)

Indicative budget bands (to be refined after discovery):

- **MVP** — CRM + AI calling + hand-off: ₹ mid-range, 8–10 weeks
- **Full platform** — all modules: ₹ premium, ~20 weeks
- **Monthly running** — Twilio, OpenAI, hosting, WhatsApp, monitoring (usage-based)

Exact figures provided in a separate commercial proposal.

19. Deliverables & Next Steps

Deliverables

- Production web app: Public website, Customer portal, Internal CRM (one codebase)
- AI telecalling engine with configurable scripts & knowledge base
- Automation workflows & human hand-off system
- PostgreSQL database with full schema, migrations, seed data
- Admin console, analytics dashboards, exports
- API/integration layer (WhatsApp, SMS, Ads, Payments)
- Documentation, deployment (Docker), training & handover

Decisions Needed From You

1. Confirm scope: full platform vs phased MVP first.
2. Confirm languages (Hindi / English / regional) and call-disclosure policy.
3. Confirm telephony: keep/port existing number(s), new Twilio numbers, or SIP trunk.
4. Provide brand assets, service areas, pricing, AMC plans, and knowledge base content.
5. Confirm integrations: which ad platforms, WhatsApp number, payment gateway, Google Business.
6. Compliance: DND/consent requirements and legal review ownership.

20. Risks & Assumptions

Risk / Assumption	Mitigation
Telecom/DND regulations	Scrub + consent flags + legal review before go-live
AI misunderstanding in regional accents	Language tuning, fallback to human, continuous transcript review
Twilio/OpenAI usage cost at 5k calls/day	Throttling, call-duration caps, monitoring, tiered concurrency
Lead data quality from uploads	Validation + dedupe + DND scrub pipeline
Customer trust in AI	Transparent disclosure + one-tap human hand-off
Integration availability (Ads/WhatsApp APIs)	Adapter pattern; start with webhooks, add APIs as available